

Brickfall Breaker Arcade - Evaluation Report

Harbor Web Development Task | Golden Oracle and model-run results

Harbor task: turing/brickfall-breaker-arcade | version 1.0.0 | prepared 12 September 2026

Scope: recorded run outcomes and task description. This document is not an independent fairness assessment or complete QC sign-off. The completed Haiku replacement is included alongside Oracle, GPT-mini and Gemini.

Headline

The golden Oracle scored 0.9583 overall and passed all 25 Functional criteria, plus every Render, Constraints and Polish criterion. Its remaining deductions were Visual. GPT-5.4 mini scored 0.6094; Claude Haiku 4.5 scored 0.1689 with both gates passed; Gemini 3.7 Flash recorded a Constraints-gated 0.0000. The empty-submission control scored 0.0000.

Title and description

Brickfall is a full-stack, deterministic brick-breaker arcade game with authenticated player profiles, ten seeded levels, power-ups, resumable SQLite-backed games, personal run records, a global leaderboard and a non-scoring mechanics lab. The short product brief is expanded by eight supplied specification files, a workbook and deterministic scenario data.

The delivered app uses vanilla HTML/CSS/JavaScript and canvas, Node.js, Express and SQLite. The runtime entry is node /app/server.js on port 3000, serving /app/public/index.html and persisting /app/brickfall.db. Both agent and separate verifier networking are public. The packaged runtime brief nevertheless requires local assets and no runtime installation; this report describes the package as supplied.

Five batched browser judge dimensions contain 40 criteria: Render 2, Constraints 2, Functional 25, Polish 5 and Visual 6. The judge is Codex with gpt-5.6-luna, temperature 0 and configured reasoning effort max. Render and Constraints are prerequisites; the final runner writes the gated score below.

if Render <= 0 or Constraints <= 0: Overall = 0
otherwise: Overall = 0.6 x Functional + 0.2 x Polish + 0.2 x Visual

Overall scores

Run	Overall	Render	Constr.	Func.	Polish	Visual	Recorded status
Oracle	0.9583	1.0000	1.0000	1.0000	1.0000	0.7917	25/25 Functional
GPT-5.4 mini (high)	0.6094	1.0000	1.0000	0.3768	1.0000	0.9167	9/25 Functional
Gemini 3.7 Flash (high)	0.0000	1.0000	0.0000	0.3986	0.2000	0.9583	9/25 Functional; gate-zero
Claude Haiku 4.5	0.1689	1.0000	1.0000	0.0870	0.0000	0.5833	2/25 Functional
No-op control	0.0000	0	0	0	0	0	graded=0; no_op=1

Binary counts are not equivalent to weighted scores. GPT-mini and Gemini each received 9/25 Functional passes, but their earned criterion weights differ; Haiku received 2/25. Visual uses 1-5 ratings normalized by the installed runner: raw 4 becomes 0.75, while raw 5 becomes 1.00. No-op dimension zeros are control outputs, not 40 independently graded failures.

Reward arithmetic

Run	Functional weight earned	Final calculation
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Run	Functional weight earned	Final calculation
Oracle	$34.5 / 34.5 = 1.0000$	$0.6 \times 1 + 0.2 \times 1 + 0.2 \times 0.7917 = 0.9583$
GPT-5.4 mini	$13 / 34.5 = 0.3768$	$0.6 \times 0.3768 + 0.2 \times 1 + 0.2 \times 0.9167 = 0.6094$
Gemini 3.7 Flash	$13.75 / 34.5 = 0.3986$	Constraints = 0; final reward = 0.0000
Claude Haiku 4.5	$3 / 34.5 = 0.0870$	$0.6 \times 0.0870 + 0.2 \times 0 + 0.2 \times 0.5833 = 0.1689$

Features and subfeatures

Area	Coverage
Sign-in and server authority	Three seeded profiles; issued bearer tokens; account-wide revocation; protected reads; account isolation; forged client identity handling.
Seed and manifest	Ten exact workbook walls, brick-type totals, constants, level names, speed limits and SHA-256 digests; Mira/Dev checkpoints and Polly history.
Ranked gameplay and progression	Fixed-step physics, paddle input, launch/pause/restart, Assist takeover, level progression, extra lives, terminal results and upward best-score persistence.
Deterministic mechanics lab	Brick types, Power relay, Multiball, Sticky catch, Last ball, Extra life and Final wall; visible simulation-step counter; practice leaves ranked state unchanged.
Persistence and coordination	Complete saved runs and terminal snapshots; refresh durability; expected revisions; stale-tab reconciliation; operation receipts and duplicate activation handling.
Interface and presentation	Keyboard/pointer/touch use, readable semantic state, visible feedback, reduced motion and six graded visual axes including the mobile layout.

Dimension / session scores

Dimension	Criteria	Budget	Role	Oracle	GPT-mini	Gemini	Haiku
Render	2	600 s	Gate	1.0000	1.0000	1.0000	1.0000
Constraints	2	600 s	Gate	1.0000	1.0000	0.0000	1.0000
Functional	25	9000 s	60%	1.0000	0.3768	0.3986	0.0870
Polish	5	900 s	20%	1.0000	1.0000	0.2000	0.0000
Visual	6	900 s	20%	0.7917	0.9167	0.9583	0.5833

All verifiers - recorded results

Entries below reproduce the exported verdicts, not a fresh regrade. Binary criteria show Pass/Fail; Visual entries show the raw five-point rating. All four evaluated applications have 40 exported criteria. A shared gate can assign Fail without independently exercising each affected criterion.

Render

Criterion (underscores shown as spaces)	Weight	Oracle	GPT-mini	Gemini	Haiku
page load and refresh	1.0	Pass	Pass	Pass	Pass
basic control interaction	1.0	Pass	Pass	Pass	Pass

Constraints

Criterion (underscores shown as spaces)	Weight	Oracle	GPT-mini	Gemini	Haiku
same origin application shell	1.0	Pass	Pass	Fail	Pass
self contained runtime and reload	1.0	Pass	Pass	Fail	Pass

Functional

Criterion (underscores shown as spaces)	Weight	Oracle	GPT-mini	Gemini	Haiku
seeded sign in and token basics	2.0	Pass	Pass	Pass	Fail
initial global leaderboard core	2.0	Pass	Pass	Pass	Pass
level manifest and constants	2.0	Pass	Fail	Pass	Fail
mira checkpoint freeze and restore	2.0	Pass	Fail	Pass	Fail
terminal finish and records	0.75	Pass	Fail	Fail	Fail
terminal receipt retry	0.5	Pass	Fail	Pass	Fail
terminal refresh durability	0.25	Pass	Fail	Pass	Fail
terminal keyboard restart	0.5	Pass	Fail	Fail	Fail
dev checkpoint threshold and progression	2.0	Pass	Fail	Fail	Fail
polly personal history and snapshots	1.5	Pass	Fail	Fail	Fail
all seeded walls and fresh runs	2.0	Pass	Pass	Fail	Fail
ranked controls pause save restore	2.0	Pass	Pass	Pass	Fail
assist and manual takeover	1.0	Pass	Pass	Fail	Fail
brick types fixed step and combo	2.0	Pass	Fail	Fail	Fail
power relay exclusivity and timer	2.0	Pass	Fail	Fail	Fail

Criterion (underscores shown as spaces)	Weight	Oracle	GPT-mini	Gemini	Haiku
multiball secondary loss	0.5	Pass	Fail	Fail	Fail
sticky capture launch and expiry	1.0	Pass	Fail	Fail	Fail
last ball life loss cleanup	0.5	Pass	Fail	Fail	Fail
extra life threshold once	1.0	Pass	Pass	Fail	Fail
final wall completion bonus	1.0	Pass	Pass	Fail	Fail
two tab revisions receipts and duplicate guard	3.0	Pass	Fail	Fail	Fail
complete session security and isolation	2.0	Pass	Fail	Pass	Fail
solid bricks do not block completion	1.0	Pass	Pass	Fail	Fail
bearer identity overrides client claims	1.0	Pass	Pass	Pass	Pass
terminal score increases best	1.0	Pass	Fail	Fail	Fail

Polish

Criterion (underscores shown as spaces)	Weight	Oracle	GPT-mini	Gemini	Haiku
sign in and focus quality	1.0	Pass	Pass	Pass	Fail
semantic game state and events	1.0	Pass	Pass	Fail	Fail
keyboard pointer touch gameplay	1.0	Pass	Pass	Fail	Fail
canvas and state readability	1.0	Pass	Pass	Fail	Fail
complete product coherence and reduced motion	1.0	Pass	Pass	Fail	Fail

Visual

Criterion (underscores shown as spaces)	Weight	Oracle	GPT-mini	Gemini	Haiku
visual typography	1.0	4/5	5/5	5/5	4/5
visual color and contrast	1.0	4/5	5/5	4/5	5/5
visual spacing and layout	1.0	4/5	4/5	5/5	3/5
visual hierarchy and scanability	1.0	4/5	5/5	5/5	3/5
visual overall craft	1.0	4/5	4/5	5/5	3/5
visual responsive consistency	1.0	5/5	5/5	5/5	2/5

Oracle - remaining Visual deductions

visual_typography - 4/5: Consistent sans typography with distinct eyebrow labels, headings, HUD values, body text, table text, and modal title. The dense telemetry strings are slightly small.

visual_color_and_contrast - 4/5: Cohesive dark navy arcade palette with cyan, violet, magenta, and yellow accents; canvas, controls, table, and modal remain readable. Muted telemetry and disabled controls are slightly low-contrast.

visual_spacing_and_layout - 4/5: Desktop gutters, stat cards, game panel, sidebar cards, controls, history list, and table are orderly with no overlap. Scrolling to the lower leaderboard leaves a notably large empty left column.

visual_hierarchy_and_scanability - 4/5: Brand, player identity, HUD, canvas, primary actions, setup, records, and leaderboard have clear grouping. Sidebar modules are somewhat uniform in prominence and telemetry is visually dense.

visual_overall_craft - 4/5: A polished, cohesive neon arcade language carries through the form, shell, canvas, HUD, history detail modal, lists, and leaderboard. Raw diagnostic telemetry and the long lower-page blank area are minor polish defects.

visual_responsive_consistency - 5/5: Desktop two-column layouts reflow cleanly into mobile stacks; stats become a two-column grid, the best score spans the row, controls and cards fit 375px, the canvas preserves its proportions, and the leaderboard remains contained without overflow.

The exported Oracle contains all 40 criteria; no dimension or Functional criterion is missing. All 34 binary criteria passed, while five Visual ratings were 4/5 and responsive consistency was 5/5. An overall score of 1.0000 is not claimed.

Failed verifiers - exported judge reasoning

These are the judge's recorded explanations. They are retained as evidence and are not presented as an independent certification of every causal interpretation.

GPT-5.4 mini (high)

Functional - 16 recorded failures

Criterion	Recorded judge explanation
level manifest and constants	Level metadata and constants matched, but the required full SHA-256 digests were not visibly present.
mira checkpoint freeze and restore	The seeded values survived reload, but Resume entered play and triggered a non-finite canvas-gradient error instead of clean frozen-resume evidence.
terminal finish and records	Mira's run did not progress to game over; no finish request or required terminal records were observed.
terminal receipt retry	No terminal finish request was captured for replay.
terminal refresh durability	No terminal record was produced, so durability was not established.
terminal keyboard restart	The required game-over state was never reached.
dev checkpoint threshold and progression	Dev's checkpoint was visible, but resumption stalled with the same canvas error and produced no progression request.
polly personal history and snapshots	The ten records were correct, but dialogs exposed only metadata rather than complete terminal snapshots.
brick types fixed step and combo	Advance produced one step and five balls rather than the required 120 steps and six-ball outcome.
power relay exclusivity and timer	The setup was correct, but Advance produced one step and incorrect score/timing instead of 120/240 ticks.
multiball secondary loss	The secondary disappeared without life loss, but the drill advanced one step with the wrong scoring/timer result.
sticky capture launch and expiry	Capture/release behavior was partly visible, but the drill advanced one or two steps rather than 120/240 with incorrect scoring.
last ball life loss cleanup	Life loss cleared the effect, but no held waiting ball appeared and the step/result evidence was incorrect.
two tab revisions receipts and duplicate guard	No visible clear control was available, so the required clear, stale-tab, receipt, reuse, and double-activation sequence was not established.
complete session security and isolation	Distinct bearer tokens and account isolation were observed, but the required rejected protected action in tab B was not established; the visible sign-out returned 200 and the tab auto-reconciled.
terminal score increases best	Dev's real resumed run remained at score 19,950 with no steps after the observed rendering errors, so no terminal score above 6,200 was reached.

Gemini 3.7 Flash (high)

Constraints - 2 recorded failures

Criterion	Recorded judge explanation
same origin application shell	The mandatory gate failed: successful sign-in automatically issued POST /api/game/start (200 OK) and rendered "Started Level 1", violating the prohibition on starting or resuming a run. Per the gate rule, scoring stopped and this criterion is 0.
self contained runtime and reload	The mandatory gate failed for the same reason before this criterion could be evaluated. Per the gate rule, this criterion is 0.

Functional - 16 recorded failures

Criterion	Recorded judge explanation
terminal finish and records	Finish, revision, leaderboard, save removal, and record values were correct, but the terminal snapshot retained one lost ball object instead of showing zero balls.
terminal keyboard restart	R was pressed on the reloaded menu, but it produced no request or state transition; the ready run had already been auto-created during reload.
dev checkpoint threshold and progression	Physics progression reached 23,050, four lives, and x2, but progress incorrectly raised best to 23,050, and Next Level reset to a new level-4 run at score 0/lives 3 instead of preserving the run.
polly personal history and snapshots	The ten seeded records matched, but inspected dialogs only contained summary fields and lacked phase, balls, drops, and brick-state snapshot data.
all seeded walls and fresh runs	All semantic walls, totals, speeds, revisions, run IDs, and operation IDs matched, but visible wall names were rendered uppercase rather than exactly matching manifest casing.
assist and manual takeover	Assist visibly enabled and simulation moved, but ArrowLeft/ArrowRight did not turn Assist off while paused.
brick types fixed step and combo	Visible Load Drill failed with createDrillState is not defined; telemetry remained at zero balls and bricks.
power relay exclusivity and timer	Power relay could not load because createDrillState was undefined; required drill telemetry was unavailable.
multiball secondary loss	Multiball could not load because createDrillState was undefined.
sticky capture launch and expiry	Sticky catch could not load because createDrillState was undefined.
last ball life loss cleanup	Last ball could not load because createDrillState was undefined.
extra life threshold once	Extra life could not load because createDrillState was undefined.
final wall completion bonus	Final wall could not load because createDrillState was undefined; no scenario bricks or completion evidence appeared.
two tab revisions receipts and duplicate guard	No visible clear-saved-run control or clear request was available, so the required empty baseline and clear replay could not be established.
solid bricks do not block completion	Final wall drill loading failed with createDrillState undefined, so solid-brick completion evidence was unavailable.

Criterion	Recorded judge explanation
terminal score increases best	Dev's actual stored best was 23,050 and the real terminal run scored 1,250; the best did not increase.

Polish - 4 recorded failures

Criterion	Recorded judge explanation
semantic game state and events	HUD, canvas summary, identity, revision, Assist, telemetry and history were readable, and launch/pause status changed. However, Brick types and Power relay Load Drill both failed with createDrillState is not defined, so the required lab outcome and event update could not be exercised.
keyboard pointer touch gameplay	Space launched and P/Escape toggled pause; pointer and keyboard moved the active paddle, and mobile action controls were large and reachable. In the ready state, keyboard movement shifted the paddle while the waiting ball stayed centered, and pointer movement was blocked by the full ready overlay, so the complete input requirement failed.
canvas and state readability	Level 1 canvas bricks had text cues such as "100" and "STRONG," and ready/playing/paused summaries matched their overlays. Required Brick types and Power relay drills could not load: telemetry remained paused/0 with Bricks State (0) and a createDrillState error.
complete product coherence and reduced motion	Sign-in, level selection, saved checkpoint resume, controls, effect timer, sync revision, lab tabs, seed digests, history, leaderboard, restart and sign-out were discoverable. The lab drill journey was unusable, and reduced-motion emulation did not shorten the observed 0.15s button transitions.

Claude Haiku 4.5

Functional - 23 recorded failures

Criterion	Recorded judge explanation
seeded sign in and token basics	Mira's successful login issued a 64-hex bearer token and protected reads worked, but the required pre-mutation Mira wrong-password submission was not established.
level manifest and constants	No visible seed-manifest view exposed the required brick totals, digests, or workbook constants.
mira checkpoint freeze and restore	Mira's initial protected state was revision 0 with activeRun null; no Resume control or required Orbit checkpoint existed.
terminal finish and records	The required Mira checkpoint was absent, so no exact finish journey, terminal request, or records were established.
terminal receipt retry	No qualifying Mira finish request or post-finish baseline was captured for replay.
terminal refresh durability	No qualifying Mira terminal record existed to verify across refresh.
terminal keyboard restart	No Mira game-over state was reached, so keyboard restart evidence was unavailable.
dev checkpoint threshold and progression	Dev's protected baseline was revision 0 with activeRun null and only levels 1-3 unlocked; the required level-3 checkpoint was absent.
polly personal history and snapshots	Polly's ten seeded records matched the required list, but the requested record-detail dialogs with terminal snapshots did not appear.

Criterion	Recorded judge explanation
all seeded walls and fresh runs	Only level 1 produced a successful start-run mutation; selecting later levels did not produce the required distinct accepted starts and revisions.
ranked controls pause save restore	A Polly run generated save requests, but exact frozen pause state, replay stability, ticks, and reload/resume equality were not established.
assist and manual takeover	Assist became visibly On, but ArrowLeft did not turn it Off immediately and gameplay values changed.
brick types fixed step and combo	Mechanics Lab showed only its heading with no visible drill chooser or Advance telemetry, so the Brick types scenario was unavailable.
power relay exclusivity and timer	The required Power relay drill could not be loaded because no visible lab chooser or drill controls were present.
multiball secondary loss	The required Multiball drill was unavailable in the visible Mechanics Lab.
sticky capture launch and expiry	The required Sticky catch drill was unavailable in the visible Mechanics Lab.
last ball life loss cleanup	The required Last ball drill was unavailable in the visible Mechanics Lab.
extra life threshold once	The required Extra life drill was unavailable in the visible Mechanics Lab.
final wall completion bonus	The required Final wall drill was unavailable in the visible Mechanics Lab.
two tab revisions receipts and duplicate guard	No visible clear operation was available, and the tabs did not provide the required independently coordinated accepted/conflict/duplicate request evidence.
complete session security and isolation	Two distinct Polly tokens were observed, but the first became invalid before tab-A sign-out and the sign-out request returned 401, so account-wide revocation and simultaneous-session evidence failed.
solid bricks do not block completion	The Final wall lab scenario could not be loaded, so surviving-solid completion evidence was unavailable.
terminal score increases best	Dev's required seeded/progressed level-4 run was absent, so no allowed terminal journey exceeded and persisted a higher best.

Polish - 5 recorded failures

Criterion	Recorded judge explanation
sign in and focus quality	Named fields and identity were present, but wrong-login submission showed no understandable in-page error.
semantic game state and events	Canvas had tabIndex -1; DOM omitted power-up/timer, keyboard guidance, telemetry, recent events, and lab outcome.
keyboard pointer touch gameplay	Space did not launch despite the canvas prompt. At 375×760, HUD and Launch were below an unscrollable viewport.
canvas and state readability	Bricks used colour-only differentiation; semantic power-up/timer state was absent and Mechanics Lab was empty.
complete product coherence and reduced motion	Core identity, history and leaderboard worked, but saved-run choice, timer, sync feedback, seed manifest and lab content were absent; level selection left "Game paused" while showing Launch. Reduced-motion transitions did shorten.

Haiku - completed replacement

The final Haiku archive contains run-05fd6dc8 / brickfall-breaker-arcade__gJZNsAw, completed without a trial exception. It scored 0.1689 with graded=1 and no_op=0. Both gates passed; Functional was 0.0870 (2/25), Polish 0.0000 and Visual 0.5833. An earlier run-35ea2a7d environment-start timeout is preserved in the repository but is not the Haiku result used in this report or the final job archive.

Run timing and reported agent usage

Run	Trial wall time	Input tokens	Output tokens	Agent cost
Oracle	25.6 min	Not reported	Not reported	Not reported
GPT-5.4 mini (high)	49.5 min	6,885,408	133,545	\$1.1803
Gemini 3.7 Flash (high)	33.8 min	7,679,345	116,523	\$1.6174
Claude Haiku 4.5	31.5 min	3,722,096	60,617	\$0.7885

Costs are the agent_result values in the exports, not a full bill including verifier charges. Trial times include setup and verification. Export timestamps are 11 September UTC (the run series spans early 12 September IST).

Grading integrity and provenance

- All five included trials, including the completed Haiku and no-op, identify the same uploaded task checksum. Oracle, GPT, Gemini and Haiku each export all five dimensions and 40 criteria; graded=1 and no_op=0 are present.
- All ten prompt/judge file hashes plus runner and reward-configuration hashes match current source for the judged trials. Oracle's package.json, server.js and public/index.html also match the current golden solution byte-for-byte.
- The task ZIP contains 32 source-matching files beneath exactly one brickfall-breaker-arcade/ wrapper. No task source, verifier definition or recorded score was changed during this submission preparation.
- The Oracle job archive includes its original no-op sibling. Run archives preserve original job/trial structure and verdict evidence; provider credentials, if encountered, are redacted only in delivery copies. Original exports remain unchanged.
- No new paid Oracle/model run, browser regrade or platform QC pass is claimed by this report. There is one measured trial per completed model; no multi-run uncertainty estimate is available.

Uploaded task checksum: 8c7bf92834e785cb53cc9c3b524c721adf74df7296c25d0049a260e83338bf8a

Task ZIP SHA-256: e41998a64346e553627ab7ce9263c129c4dcaad66bec8c45aaff71ab887c8c1a

Delivery contents and source index

- CASE-STUDY-brickfall-breaker-arcade.docx
- EVAL-REPORT-brickfall-breaker-arcade.docx
- brickfall-breaker-arcade-claude-haiku-4.5-job-directory.zip
- brickfall-breaker-arcade-gemini-3.7-flash-job-directory.zip
- brickfall-breaker-arcade-gpt-5.4-mini-high-job-directory.zip
- brickfall-breaker-arcade-oracle-job-directory.zip
- brickfall-breaker-arcade.zip

Oracle: run-68a1687d-379a-4722-8813-84fe4b74ccc6 / brickfall-breaker-arcade__4ZZo7QK

GPT-5.4 mini (high): run-fa9adc6b-4e7b-47ba-a8dd-f92a2dda2968 / brickfall-breaker-arcade__gWKVQwC

Gemini 3.7 Flash (high): run-82ef5e39-356b-40f3-9939-838f81bc04ce / brickfall-breaker-arcade__g7SDSC3

Claude Haiku 4.5: run-05fd6dc8-b477-4dae-8b8c-90b06fef2667 / brickfall-breaker-arcade__gJZNsAw

Evidence is under each job archive's original trial directory: result.json; verifier/reward.json; verifier/reward-details.json; verifier/prompt-provenance.json; test output; and available app/agent artifacts. The Haiku archive contains its completed replacement run. JSON files are authoritative for recorded scores.

Current standing

Oracle Functional is 1.0000, and its overall 0.9583 is above 0.95. GPT-5.4 mini's recorded 0.6094 lies within the repository's documented 0.1-0.7 keeper band. Haiku's completed score is 0.1689. Gemini's 0.0000 is a gate-zero result, not an independent measure of all its product behavior. All requested run exports are now included. This is a results handoff, not a declaration that every platform requirement has been independently certified.